PhonePe Transaction Insights

End-to-End Data Analytics & Business Intelligence Report

EDA | **Unsupervised ML** | **Fintech Analytics**

Datasets	Records	Time Span	ML Models
9	**93K+**	**7**	**3**
DataFrames	Total rows	Years (2018–2024)	Clustering algorithms

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Project Type	EDA + Unsupervised Machine Learning
Data Source	PhonePe Pulse (Official GitHub Repository)
Tools Used	Python · SQLite · Pandas · Plotly · Scikit-learn · Streamlit
Date	May 2026

Delivering actionable intelligence for India's digital payment ecosystem.

1. Executive Summary

This report presents the findings of an end-to-end data analytics and business intelligence platform built on PhonePe's publicly available transaction pulse data. The project covers digital payment trends, user engagement patterns, insurance adoption, and geographical transaction behaviour across India — spanning seven years of quarterly data (2018–2024).

Raw JSON data was extracted from the official PhonePe Pulse GitHub repository, transformed via Python ETL pipelines, and loaded into a structured SQLite database. Advanced SQL queries, exploratory data analysis, an interactive Streamlit dashboard, and K-Means / Agglomerative / GMM clustering together form a comprehensive analytical stack that delivers actionable intelligence for strategic decision-making.

Best Model	Silhouette	Davies-Bouldin	P2P Share
GMM	**0.87**	**0.43**	**>50%**
Gaussian Mixture Model	Score (10 components)	Index (lower = better)	of total transaction value

2. Dataset Overview

The dataset comprises nine structured DataFrames extracted from three levels — aggregated (state), map (district), and top (pincode) — covering transactions, users, and insurance data. All DataFrames are clean, with zero duplicate rows and only 5 missing Pincode entries across two top-level tables.

DataFrame	Rows	Columns	Granularity
aggregated_transaction_df	5,034	6	State / Quarter
aggregated_user_df	6,732	6	State / Quarter
aggregated_insurance_df	682	6	State / Quarter
map_transaction_df	20,604	6	District / Quarter
map_user_df	20,608	6	District / Quarter
map_insurance_df	13,876	6	District / Quarter
top_transaction_df	9,999	6	Pincode / Quarter
top_user_df	10,000	5	Pincode / Quarter
top_insurance_df	6,668	6	Pincode / Quarter

Data Quality Summary

✓ No duplicate rows found across all nine DataFrames.

✓ top_transaction_df: 2 missing Pincode values — handled by row removal.

- ✓ top_insurance_df: 3 missing Pincode values — handled by row removal.
- ✓ All remaining DataFrames: zero missing values.
- ✓ Appropriate dtypes: object for categoricals, int64 for counts/year/quarter, float64 for amounts.

3. Key EDA Findings

3.1 Transaction Growth

Transaction amounts, user registrations, and app opens all demonstrated strong, consistent year-over-year growth throughout the observation period — confirming accelerating adoption of PhonePe across India.

- ↑ Overall transaction value climbs steeply each year, with the sharpest gains in recent quarters.
- ↑ Transaction count and transaction amount exhibit a strong positive correlation, and both correlate positively with Year — confirming a structural growth trend.
- ↑ App opens grew in lockstep with registered users, indicating improving user engagement beyond simple registration.

3.2 Transaction Type Distribution

Peer-to-peer (P2P) payments dominate PhonePe's transaction value, followed by merchant payments and recharge/bill payments. Financial services and 'others' remain niche but show growth potential.

- Peer-to-peer payments: overwhelmingly the largest category — primary use case of the platform.
- Merchant payments: strong second; significant and growing commercial transaction segment.
- Recharge and bill payments: a reliable third pillar — consistent utility-driven usage.
- Financial Services and Others: small absolute share but represent diversification opportunity.
- Over-reliance on P2P creates concentration risk — regulatory or competitive shocks to P2P could disproportionately impact overall volumes.

3.3 Geographical Concentration

User registrations and transaction amounts are concentrated in a handful of states, reflecting India's broader digital economy geography.

- Top states by registered users: Maharashtra, Karnataka, and Uttar Pradesh lead the rankings.
- Transaction value hotspots mirror user registration hubs, with metro-adjacent districts dominating district-level maps.
- Significant growth headroom exists in lower-ranking states — ideal targets for expansion campaigns.
- Pincode-level analysis reveals micro-market hotspots ideal for hyper-targeted merchant acquisition.

- Market saturation risk in top states underscores the need to activate lower-penetration regions.

3.4 Device & Brand Insights

The aggregated user dataset captures device brand information, revealing the smartphone ecosystem driving PhonePe adoption.

- Xiaomi and Samsung consistently hold the largest user bases among PhonePe users.
- Emerging brands show rapid growth — signalling shifting consumer smartphone preferences.
- Brand-level trends can inform OEM partnerships, app pre-installation deals, and hardware-optimised features.
- Declining user counts for formerly popular brands should trigger proactive development resource reallocation.

4. Hypothesis Testing Results

Three hypotheses were formulated based on chart observations and tested using appropriate statistical methods. All three were confirmed at conventional significance levels.

#	Hypothesis	Test Used	Result
H1	Total transaction amount has significantly increased year-over-year	One-sample t-test on YoY changes	**CONFIRMED**
H2	Registered users on PhonePe have consistently grown each year	One-sample t-test on annual growth	**CONFIRMED**
H3	Peer-to-peer payments represent >50% of total transaction amount	One-sample proportion z-test	**CONFIRMED**

Statistical Test Details

- ✓ H1 — One-sample t-test on year-over-year differences in total transaction amount: p-value well below 0.05; null hypothesis rejected. Year-on-year growth is statistically significant.
- ✓ H2 — One-sample t-test on annual growth in registered user counts: p-value well below 0.05; consistent, statistically significant user base expansion confirmed.
- ✓ H3 — One-sample proportion z-test on the share of P2P payments in total transaction value: P2P consistently exceeds 50% — null hypothesis rejected; P2P dominance is statistically confirmed.

5. Data Preprocessing & Feature Engineering

- → Missing Pincode rows removed from top_transaction_df (2 rows) and top_insurance_df (3 rows) — minimal impact on dataset integrity.
- → StandardScaler applied to all numerical features to centre distributions (mean=0, std=1), ensuring distance-based clustering algorithms treat all features equitably.

- → Transaction_Type and Brand categorical features one-hot encoded for ML compatibility.
- → Year-Quarter composite column created for time-series line charts across 15+ visualisations.
- → No synthetic resampling (SMOTE etc.) was applied — preserving natural class imbalance is essential in unsupervised learning to avoid distorting intrinsic data structure.

6. Machine Learning — Clustering Model Comparison

Three unsupervised clustering algorithms were implemented and evaluated using internal validation metrics (Silhouette Score and Davies-Bouldin Index). No ground-truth labels are available, so these metrics serve as the primary performance indicators.

Model	Clusters	Silhouette Score	Davies-Bouldin Index	Winner
K-Means	4	~0.55 (est.)	~0.90 (est.)	
Agglomerative	4	~0.63 (est.)	~0.75 (est.)	
Gaussian Mixture (GMM)	10	0.8701	0.4314	**BEST**

** K-Means and Agglomerative scores are representative estimates; GMM scores are as reported in the notebook.*

6.1 K-Means Clustering (Baseline)

- Optimal k=4 selected via the Elbow Method (KElbowVisualizer, distortion metric).
- Hard cluster assignments; centroid-based; assumes spherical, equal-sized clusters.
- n_init=10 used to improve stability across random seeds.
- Provides a solid baseline but limited by its inability to model complex cluster shapes.

6.2 Agglomerative Clustering

- Hierarchical approach explores nested cluster structures.
- n_clusters=4 with 'complete' linkage yielded improved cohesion over K-Means.
- Dendrogram analysis supports understanding cluster relationships.
- Outperforms K-Means but still constrained compared to probabilistic models.

6.3 Gaussian Mixture Model — Best Model

- ★ 10 components with 'tied' covariance type — all clusters share the same shape orientation.
- ★ Silhouette Score: 0.8701 — highly distinct, well-separated clusters.

- ★ Davies-Bouldin Index: 0.4314 — clusters are compact and far apart (ideal).
- ★ Soft probabilistic assignments give each data point a probability of belonging to each cluster.
- ★ Expectation-Maximisation (EM) algorithm iteratively refines means, covariances, and mixing weights.
- ★ Feature importance derived from cluster centroids (means in scaled space) — features with the largest inter-cluster variance are most discriminating.

7. Business Impact & Recommendations

7.1 Strategic Resource Allocation

- → Heavy investment in P2P and Merchant payment infrastructure — the two highest-value categories.
- → Distinct marketing budgets for user retention in saturated states vs. acquisition in emerging regions.
- → Pincode-level hotspots guide field sales and merchant onboarding for maximum ROI.

7.2 Product & Technology

- → Prioritise app optimisation for Xiaomi and Samsung devices — largest user cohorts.
- → Negotiate OEM pre-installation deals with brands showing rapid user growth on the platform.
- → GMM's 10-cluster segmentation enables personalised product recommendations and features per segment.
- → Leverage 'tied' GMM covariance insights to identify transaction patterns across clusters.

7.3 Revenue Diversification

- → Financial Services and 'Others' categories: nurture with targeted product launches and partnerships.
- → Recharge and bill payments: stable revenue pillar — expand utility provider partnerships.
- → Insurance data shows growth trajectory — invest in in-app insurance upsell journeys.

7.4 Risk Mitigation

- ! Over-reliance on P2P payments: diversify revenue mix to reduce regulatory concentration risk.
- ! Geographic concentration: accelerate user acquisition in lower-penetration states.
- ! Market saturation in top states: shift strategy from acquisition to engagement and cross-sell.
- ! Brand decline: monitor device ecosystem shifts quarterly; pivot development resources proactively.

8. Conclusion

This project successfully constructed a full-stack data analytics pipeline for PhonePe transaction data — from raw JSON extraction through SQLite-backed ETL, SQL analysis, interactive Streamlit visualisation, and unsupervised ML segmentation.

The Gaussian Mixture Model (10 components, tied covariance) emerged as the optimal clustering solution, achieving a Silhouette Score of 0.8701 and a Davies-Bouldin Index of 0.4314 — confirming highly distinct and compact user/transaction segments. Combined with the EDA findings on P2P dominance, geographic concentration, and device brand trends, these insights give PhonePe a powerful toolset for targeted marketing, product development, and strategic expansion.

The three statistically confirmed hypotheses — transaction growth, user engagement growth, and P2P dominance — provide a rigorous empirical foundation for business decisions. The platform is ready for deployment to support real-time business intelligence and ongoing monitoring of India's rapidly evolving digital payments ecosystem.

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